Pattern Recognition 153 (2024) 110506

Contents lists available at ScienceDirect

Pattern Recognition

journal homepage: www.elsevier.com/locate/pr

Rebalancing network with knowledge stability for class incremental learning

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ARTICLE INFO

Keywords:
Class incremental learning
Catastrophic forgetting
Class imbalance
Proxy-based metric learning
Knowledge distillation

ABSTRACT

Class incremental learning (CIL) has been proposed to solve the problem of learning to classify new classes while maintaining the performance on old classes. A typical strategy is to update the old classification model with entire new class data and a few old exemplars, which face serious performance degradation on old classes. The main reasons come down to class imbalance between old and new classes along with catastrophic forgetting towards old classes. Most existing CIL methods have proposed solving the above two issues in classification space, ignoring their adverse effects of overlapping between old and new classes and confusion among old classes in feature space. In this paper, we propose a rebalancing network with knowledge stability (RNKS), aiming to adequately retain the model performance on old classes in CIL by solving class imbalance and catastrophic forgetting in feature and classification space simultaneously. In detail, the proposed RNKS mainly consists of multi-proxies rebalancing (MPR) and hybrid knowledge distillation (HKD). MPR, focusing on class imbalance, employs multi-proxies metric learning to decrease the feature overlapping between old and new classes, together with balanced data sampling to correct the skewed decision boundary. HKD, coping with catastrophic forgetting, encourages the updated model to reproduce identical feature topologies and predictions of old classes as the old model via feature relation-based and response-based distillations. Experiments on CIFAR-100 and ILSVRC datasets demonstrate the effectiveness of this work against the state-of-the-art approaches.

1. Introduction

Traditional object classification algorithms perform model training based on prepared training sets, where the number of categories is fixed in advance [1,2]. In contrast, new categories may continue to appear in real-world scenarios, and the need to additionally learn to classify new categories is prevalent [3]. For instance, an autonomous driving platform may encounter various new objects when traveling to different regions. In order to ensure its validity in multifarious scenarios, learning to classify new objects without losing the performance on old objects is meaningful. To achieve this goal, it is necessary for traditional classification algorithms to constantly retrain the model with all data from old and new classes. However, retraining the model every time a new object is observed brings huge storage pressure and time consumption [4,5], making the model updating process inefficient.

Aiming to enhance the updating efficiency of classification models in real-world scenarios, class incremental learning (CIL) is proposed. Existing CIL methods usually preserve a few old exemplars to update the old model together with new class data (as shown in Fig. 1), which is called rehearsal strategy [6]. Such idea to enhance model efficiency can be seen in researches from other fields [7,8], which selects representative samples that benefit classification to train the model, aiming to construct a real-time classification model with high accuracy and fast convergence speed. However, differing from their applications in the fixed classification task, the CIL process of co-updating model with a few old exemplars and newly added class data faces significant performance degradation on old classes. Specifically, the reason mainly comes down to two points: (1) class imbalance between old and new class; (2) catastrophic forgetting towards old classes. Firstly, there exists a huge quantitative gap between old exemplars and new data, such class imbalance induces the favoritism of the new model to misclassify old class data into new classes [9]. In addition, stored old exemplars are insufficient to represent entire old data, so that the learned discrimination among old classes will be lost in the update process, referred to as catastrophic forgetting [10].

Considerable research efforts have been devoted to overcome the above performance degradation problem from the following two aspects.

(1) How to correct the imbalanced learning between old and new classes

One approved consequence of imbalanced learning is that the learned classification boundary is skewed towards new classes, resulting in strong favoritism of misclassifying old class data into new

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https://doi.org/10.1016/j.patcog.2024.110506
Received 18 August 2023; Received in revised form 10 November 2023; Accepted 15 April 2024
Available online 18 April 2024

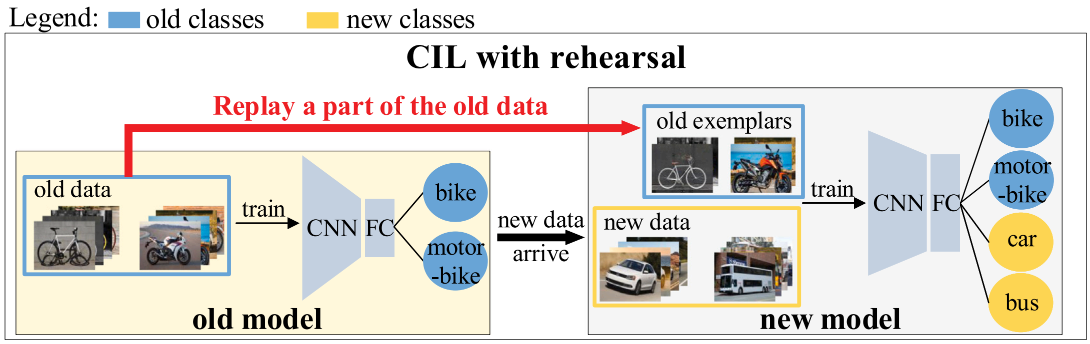

Fig. 1. Illustration of CIL with rehearsal strategy. Firstly, the old model is trained with old class data. Afterwards,some exemplars from old classes are stored in limited memory. When new class data arrive, the new model is first constructed by expanding the classification node of the old model. Then, the stored exemplars are replayed and used with new data to train the new model, so that it can correctly identify all classes appeared so far.

classes [9,11]. Based on such perception, various classification-level solutions such as EEIL [12], BIC [11], WA [1] and OCLvCV [13] have been proposed. However, class imbalance causes not only the skewed classification layer, but also the feature overlapping between old and new classes. There are a few works [14,15] aim to solve the above overlapping by modifying the feature distribution of old classes. However, due to their biased and limited modification on old classes, none of them can properly represent the feature distribution of old classes after more new classes are added. How to enhance the separability between old and new classes in both classification and feature space throughout CIL processes should be considered.

(2) How to retain the old knowledge to alleviate catastrophic forgetting

Old knowledge-retaining researches usually introduce extra constraints in the loss function to consolidate old knowledge in the incremental update [3], called regularization strategy. Among these researches, knowledge distillation [16] is most commonly used in existing CIL methods, which transfers knowledge among old classes learned by entire old data from the old model to the new model, and thus alleviates catastrophic forgetting. Lots of existing methods such as DMC [17], COIL [18] and DCV [19] only impose distillation constraints on classification logits of the model, wasting the old knowledge contained in the feature space. Recently, a few works such as AFC [20], LUCIR [9] and PRD [21] propose feature-level distillations. Some of them [20,21] ignores the crucial knowledge of discrimination among old class features, while the other one [9] retains the crucial knowledge by sacrificing the model flexibility for learning new classes. How to reasonably retain all crucial old knowledge should be considered.

Considering the insufficiency of most existing CIL methods that only mitigates class imbalance and catastrophic forgetting in classification space, along with the shortcomings of a few techniques designed for feature space, we propose a novel CIL framework called rebalancing network with knowledge stability (RNKS). On the basis of rehearsal-based CIL, the motivation of our RNKS is to sufficiently and reasonably alleviate class imbalance and catastrophic forgetting in both classification space and feature space, thus retaining the performance on old classes during CIL processes. In detail, the basic classification network of RNKS consists of a backbone CNN (convolutional neural network) for feature extraction and a single FC (fully connected) layer for classification. To alleviate class imbalance, we introduce a multi-proxies rebalancing (MPR) module in RNKS. In detail, a separable feature learning branch parallel with the classification branch is designed, which cooperates with a multi-proxies metric loss to enhance the distinguishability of old and new class features extracted by the backbone CNN. In the classification branch, a cross-entropy loss that cooperates with balanced sampled data is exploited to learn an unbiased classification layer. Additionally, to mitigate catastrophic forgetting, we further devise a hybrid knowledge distillation (HKD) module to achieve the stability of learned knowledge among old classes. Specifically, two distillation losses are leveraged to transfer the feature-level and classification-level discriminative knowledge from the old model to the new model, respectively. At feature-level, a novel feature relation-based distillation loss is employed to flexibly transfer the important old knowledge of intra-class compact and inter-class separable feature topologies. At classification-level, a response-based distillation loss is used to transfer the accurate predictions. Overall, 4 losses are concerned during each CIL process of our RNKS, including multi-proxies metric loss and cross-entropy loss for in MPR, along with feature relation-based and response-based distillation losses in HKD.

With the above techniques, our RNKS can efficiently accumulate the classification knowledge in dynamic scenarios, enhancing the cognitive abilities of real-world application platforms such as autonomous driving and robots. The contributions of this work are outlined as follows:

1. We design an MPR module to solve class imbalance, which first tackles this issue in CIL from perspectives of both classification space and feature space. Especially in feature learning, a multi-proxies metric loss is employed. With rigorous derivation, it is proved to be robust for enhancing the feature separability of old and new classes under class imbalance condition. By further using balanced sampled data to learn an unbiased classifier, the effects of class imbalance can be corrected.
2. We devise an HKD module for alleviating catastrophic forgetting, which innovatively transfers discriminative knowledge of well-represented feature topologies and accurate predictions from the old model to the new model, ensuring the stability of learned knowledge so that confusions among old classes in CIL process can be mitigated.
3. Experiments on CIFAR-100 and ILSVRC benchmarks certify the validity of our approach, which outperforms the state-of-the-art CIL approaches.

The rest of this article is organized as follows. We first discuss the related work in Section 2 and then explain the methodology in Section 3. In Section 4, we report the experimental analyses. Finally, we draw a conclusion in Section 5.

2. Related work

2.1. Rehearsal

In earlier researches, a naive CIL method is to expand the classification layer of the old model and fine-tune it on new data [22], suffering from the risk of totally losing the performance on old classes [23]. Then, the rehearsal strategy is first proposed by iCaRL [6], which keeps a few old exemplars nearing the class mean feature to update the model together with new data. Since then, various rehearsal researches have been proposed, which focus on exemplars around the decision boundary [24,25], concentrate on diversity [26], or consider discriminability and diversity comprehensively [27]. Since the rehearsal strategy in [6] is widely used as benchmarks in existing CIL methods, we also adopt their strategy in this paper.

2.2. Class imbalance correction

To mitigate the class imbalance between old exemplars and new class data, multifarious researches have been proposed. A common observation is that the magnitudes of both the weights and the biases of the classification layer have strong favoritism towards new classes. To rebalance such favoritism, WA [1] rescales the output logits in the testing phase. EEIL [12] and BIC [11] add an extra balanced fine-tuning stage after updating model with imbalanced data. OCLvCV [13] decouples the predictions of old and new classes, avoiding the biased logits towards new classes in a single-head classifier. Nevertheless, the above methods ignore the effects of class imbalance in feature space, which causes feature overlapping between old and new classes. SDC [14] proposes to use nearest class mean (NCM) classifier in feature space, while the feature overlapping is solved by rectifying the position of old class mean vectors via new class features. Since such a process is decoupled with old class features, whether the rectified old class mean vectors can represent old class features is dubious. Recently, CwD [15] observes the importance of compressing the feature distribution of old classes and adds a regularization loss in the initial phase, but cannot prevent the overlapping between old and new classes in multiple incremental learning phases. Similarly but differently, in our method, the learning of intra-class compactness and inter-class separability among old and new classes is performed in each learning phase, alleviating the feature overlapping throughout the CIL process.

2.3. Regularization

Researches on regularization aim to alleviate catastrophic forgetting by imposing constraints related to old knowledge when learning new classes. Among them, knowledge distillation [19] is commonly used. LWF [9] is a pioneering work that first constrains the new model to reproduce the predictions of the old model on old class data. DMC [28] trains two separate networks for old and new classes and then combines them via a double distillation loss. COIL [29] distills the knowledge of semantic relationships between old and new classes to benefit new class learning and old knowledge retaining simultaneously. DCV [30] employs a dynamic correction vector to correct the biased output of the old model in the distillation loss. However, narrowly transferring old knowledge from the classification space is insufficient. AFC [31] chooses to further impose distillation constraints on the important feature maps, neglecting the crucial learned knowledge of discrimination among old class features. To retain such knowledge, LUCIR [32] adds a strong constraint of the cosine similarity on feature vectors, losing the model flexibility for learning new classes. PRD [33] designs a relation distillation, which flexibly distills the relation between old class centroid and new class features. Without the restraint between old class centroid and old class features, the learned old class discrimination will also be destroyed. In our method, the relation distillation is also adopted on account of the model flexibility. But differently, we propose to retain the relation of high intra-class similarity and low inter-class similarity about old class features in the old model, so that the learned discrimination among old classes can be adequately protected.

3. Methodology

3.1. Problem formulation

CIL aims to learn new classes from streaming data while maintaining the performance on old classes learned before. Given a sequence of T learning phases

$$T = \left[\left(\mathbf{C}^1, \mathbf{D}^1\right), \left(\mathbf{C}^2, \mathbf{D}^2\right), \ldots, \left(\mathbf{C}^T, \mathbf{D}^T\right)\right] \tag{1}$$

each learning phase t exists a set of newly observed classes $\mathbf{C}^t$ and its training dataset $\mathbf{D}^t$. $\mathbf{C}^t = \left\{c_1^t, c_2^t, \ldots, c_{n_t}^t\right\}$ contains n_t new classes which is disjoint with other learning phases (i.e., $\mathbf{C}^i \cap \mathbf{C}^j = \emptyset \text{if} i \neq j$). $\mathbf{D}^t = \left\{\left(\mathbf{x}_1^t, y_1^t\right), \left(\mathbf{x}_2^t, y_2^t\right), \ldots, \left(\mathbf{x}_{m_t}^t, y_{m_t}^t\right)\right\}$ contains m_t training data belonging to $\mathbf{C}^t$, where $\mathbf{x}_i^t$ represents ith sample in $\mathbf{D}^t$ and y_i^t denotes its corresponding ground-truth label. The initial model M_1 trained on $\mathbf{D}^1$ is expected to gradually recognize more classes by learning $\left(\mathbf{C}^t, \mathbf{D}^t\right)$ $(t \in \{2, \ldots, T\})$ sequentially. Specifically, in each phase t, the old model M_{t-1} is able to classify the learned classes $\bigcup_{k=1}^{t-1} \mathbf{C}^k$ (simply denoted as $\mathbf{C}^o$), an incremental learning phase is defined as constructing M_t to correctly classify $\bigcup_{k=1}^{t} \mathbf{C}^k$ by updating M_{t-1} with $\mathbf{D}^t$ and a few stored old exemplars $\left(\bigcup_{i=1}^{t-1} \mathbf{D}^i\right)_{stored}$ (simply denoted as $\mathbf{D}^o$) obtained in previous phases.

In the following section, we will introduce the details of proposed RNKS in a single incremental learning phase t for easier understanding.

3.2. The overall framework

The overall architecture of our RNKS is shown in Fig. 2, which follows a framework similar to traditional student–teacher knowledge distillation, transferring knowledge from teacher model to student model. Notably, comparing to traditional distillation used in the model compression field, the characters of our distillation framework for CIL is different. Concretely, the teacher model in CIL is the frozen old classification model M_{t-1} for supplying old knowledge, while the student model in CIL is the learnable new model M_t for incrementally learning new classes. During the training process of M_t, four modules are involved, including data sampling, the backbone CNN, the MPR and the HKD, which are briefly introduced as follows.

(1) Data sampling: To better optimize feature space and classification space under class imbalance condition, we adopt different sampling strategies to generate batches of training data. In detail, given the training dataset containing $\mathbf{D}^o$ and $\mathbf{D}^t$, it is sampled into sets of $\mathbf{D}_{rad}$ and $\mathbf{D}_{bal}$ via random sampling and class balanced sampling, respectively. Given the probability p_j of sampling a data from class j: $p_j = n_j^q / \sum_{k=1}^{m+n} n_k^q$, where $q \in [0, 1]$ and n_k represents the number of data belonging to class k. For random sampling, $q = 1$, which retains original data distribution and benefits well-represented feature learning. For class balanced sampling, $q = 0$, which ensures data balance among all classes and helps to eliminate the favoritism towards new classes in classifier learning. When training M_t, a set of $\mathbf{D}_{rad}$ and $\mathbf{D}_{bal}$ is viewed as a mini-batch for the gradient optimization.

(2) Backbone CNN: The backbone CNN is a ResNet [34], which aims to extract deep feature for each sample. When the $\mathbf{D}_{rad}$ and $\mathbf{D}_{bal}$ are fed into M_t as inputs, they will be transformed as corresponding features $\hat{\mathbf{Z}}_{rad} \in \mathbb{R}^{(N/2)\times K}$ and $\hat{\mathbf{Z}}_{bal} \in \mathbb{R}^{(N/2)\times K}$ via the backbone CNN and the global average pooling (GAP), where N denotes the number of samples contained in a mini-batch and K implies the dimension of features obtained by the backbone CNN and GAP.

(3) MPR: The MPR module is used to solve class imbalance. It consists of two branches shared by the backbone CNN, including the upper branch for unbiased classifier learning and the lower one for separable feature learning. Specifically, $\hat{\mathbf{Z}}_{rad}$ are input into separable feature learning branch, and each of them is embedded as $\hat{\mathbf{r}} \in \mathbb{R}^E$ through an embedding function $g_e(\cdot)$ for the optimization of multi-proxies metric loss L_{mp}. In addition, $\hat{\mathbf{Z}}_{bal}$ are input into unbiased classifier learning branch, where each of them is mapped as classification logits $\hat{\mathbf{o}} \in \mathbb{R}^{m+n}$ via the FC layer in Fig. 2, for the optimization of cross entropy loss L_{ce}.

(4) HKD: The HKD module is used to alleviate catastrophic forgetting. When optimizing M_t, $\mathbf{D}_{rad}$ and $\mathbf{D}_{bal}$ are input into the old model M_{t-1} and the new model M_t simultaneously. Following the data sampling strategies for MPR, the output features $\mathbf{Z}_{rad} \in \mathbb{R}^{(N/2)\times K}$ and logits $\mathbf{o} \in \mathbb{R}^m$ obtained by M_{t-1} will be used as supervised information to constrain the learning of M_t. Specifically, the feature $\mathbf{Z}_{rad}$ and $\hat{\mathbf{Z}}_{rad}$ are used for the optimization of feature relation-based distillation loss L_{frd}

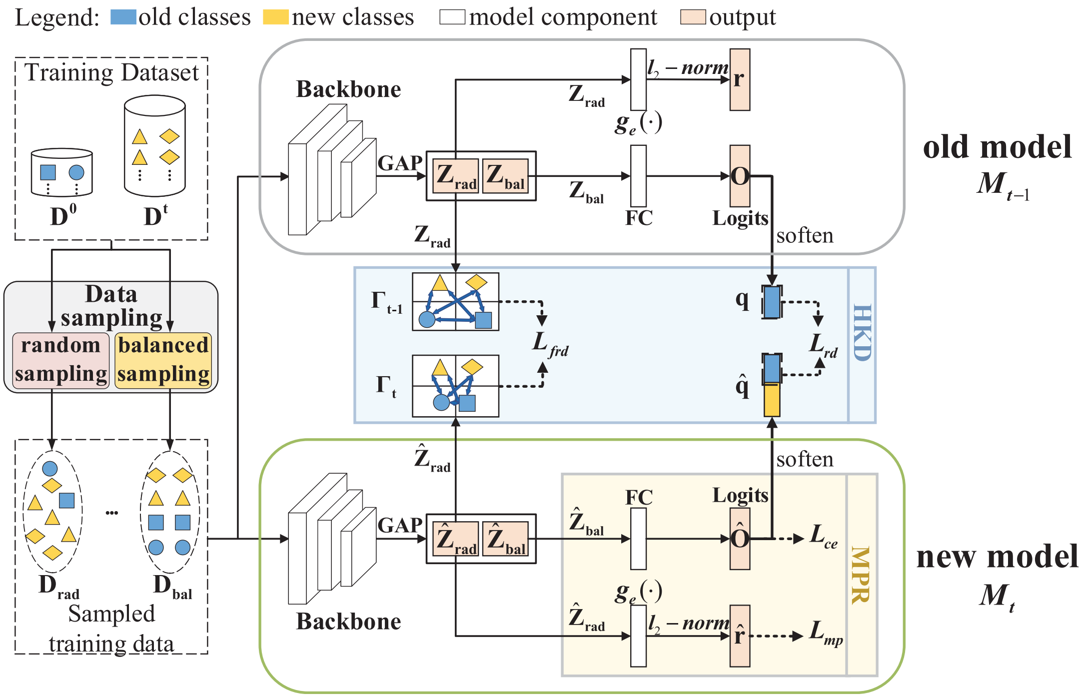

Fig. 2. Framework of our proposed RNKS in the tth incremental learning phase.

based on their topologies Γ_{t-1} and Γ_t, and the softened logits $\mathbf{q}$ from $\mathbf{o}$ along with the softened logits related to old classes in $\hat{\mathbf{q}}$ from $\hat{\mathbf{o}}$ are used for the optimization of response-based distillation loss L_{rd}.

Among above four modules, the details of the key module MPR and HKD in our framework are introduced concretely in Sections 3.3 and 3.4, respectively.

3.3. Multi-proxies rebalancing (MPR)

Owing to the class imbalance between stored old exemplars and new data, the CIL model focuses mainly on learning new classes in the updating process, which leads to serious favoritism towards new classes.

Given an input sample $\mathbf{x}_i$ belonging to an old class, its feature $\hat{\mathbf{z}}_i$ is fed into the classification layer. Then, its classification logits can be obtained:

$$\hat{\mathbf{o}}\left(\mathbf{x}_i\right) = \mathbf{W}^T \cdot \hat{\mathbf{z}}_i + \mathbf{b} = \left[\mathbf{w}_1\hat{\mathbf{z}}_i + b_1, \mathbf{w}_2\hat{\mathbf{z}}_i + b_2, \dots, \mathbf{w}_{m+n}\hat{\mathbf{z}}_i + b_{m+n}\right] \tag{2}$$

where $\mathbf{W} = \left[\mathbf{w}_1, \dots, \mathbf{w}_{m+n}\right] \in \mathbb{R}^{K\times(m+n)}$ and $\mathbf{b} \in \mathbb{R}^{m+n}$ are the weight and bias term of the classification layer, m and n denotes the number of classes in old class set $\mathbf{C}^o$ and new class set $\mathbf{C}^t$, respectively. Affected by imbalanced learning that mainly focuses on new classes while neglecting old classes, adverse consequences will occur: (1) confused feature space: $\hat{\mathbf{z}}_i$ is seriously overlapped with features of new classes; (2) biased classification layer: both the L2 norm of new class weights $(\mathbf{w}_{m+1}, \dots, \mathbf{w}_{m+n})$ and the values of new class biases $(b_{m+1}, \dots, b_{m+n})$ are dramatically larger than those of old class weights $(\mathbf{w}_1, \dots, \mathbf{w}_m)$ and biases $(b_1, \dots, b_m)$, so that $\mathbf{x}_i$ tends to be misclassified as new classes with a high probability.

To figure out the above issues, we propose MPR, which is composed of a separable feature learning (SFL) branch and an unbiased classifier learning (UCL) branch, as can be seen in Fig. 3. The two learning branches shared by the backbone CNN are learned based on different sampled data and possess different characters in class imbalance alleviation. Concretely, the purpose of such learning strategies for the two branches is outlined as follows: (1) the $\mathbf{D}_{bal}$ obtained by balanced sampling is used for UCL, which ensures the classification layer is learned by class balanced data, promoting the weight and bias terms to be unbiased; (2) the $\mathbf{D}_{rad}$ acquired by random sampling is used for SFL, which maintains the original data distribution. Compared with $\mathbf{D}_{bal}$, $\mathbf{D}_{rad}$ is more suitable for the optimization of backbone CNN, since the old class data is oversampled in balanced sampling, which makes the backbone CNN prone to overfitting. The details of above two branches will be introduced hereinafter.

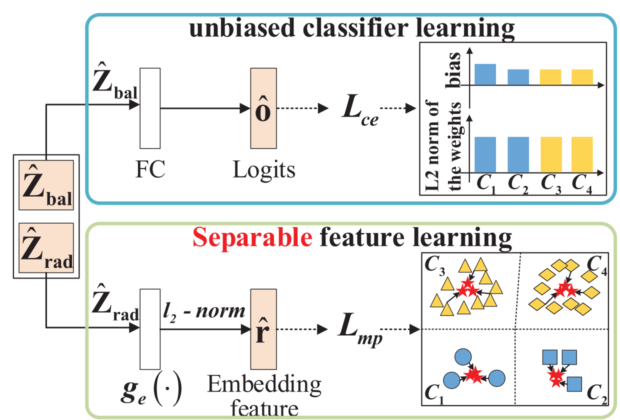

Fig. 3. Schematic diagram of multi-proxies rebalancing module. The proxies for multi-proxies metric learning are marked as stars.

The SFL branch locates the lower one of MPR shown in Fig. 3, leveraging multi-proxies metric mechanism to learn distinguishable features. To better enhance the quality of features extracted by the backbone, the feature $\hat{\mathbf{z}}_{rad}$ of the sample from $\mathbf{D}_{rad}$ is mapped into a more suitable embedding space for metric learning via $g_e(\cdot)$ [28]. Then, the l_2-normalized embedding feature $\hat{\mathbf{r}}$ is used to optimize L_{mp}, which assigns multiple learnable proxies for each class, and maximizes the similarity between samples and its corresponding proxies while minimizing that between samples and proxies of other classes. For an input sample $\mathbf{x}_i$, its embedding feature can be represented as $\hat{\mathbf{r}}_i$. Then, the multi-proxies metric loss function is denoted as follows:

$$L_{mp}\left(\hat{\mathbf{r}}_i\right) = -\log\left[\exp\left(\sum_{s=1}^{S}\hat{\mathbf{r}}_i\cdot\mathbf{p}_{y_i}^s\right)\Big/\sum_{j=1}^{m+n}\exp\left(\sum_{s=1}^{S}\hat{\mathbf{r}}_i\cdot\mathbf{p}_j^s\right)\right] \tag{3}$$

where y_i is the label of $\mathbf{x}_i$, S denotes the number of proxies for each class, $\mathbf{p}_j^s$ is the sth proxy of class j, $\mathbf{p}_{y_i}^s$ is the sth proxy of class y_i. Both $\hat{\mathbf{r}}_i$ and proxies have been normalized to ensure that their inner product can be used as cosine similarity. In particular, compared with standard cross entropy loss which easily makes the optimization of classes with sufficient samples overwhelm that of classes with few samples [29], L_{mp} is insensitive to class imbalance. Specifically, when optimizing the backbone CNN via L_{mp}, the separability between old and new classes is not only learned by the limited old exemplars and proxies of new classes, but also by the sufficient new class data and proxies of old classes, such bi-directional separability optimization eliminates the imbalanced learning in feature space.

Moreover, it should be noticed that our multi-proxies metric loss is an extension of the Proxy-NCA loss (i.e. $S = 1$) [30]. However, the effect of multi-proxies setting in proxy-based metric learning is an unexplored issue. In this paper, we first observe that different settings of S brings different power in proxy-based metric loss, resulting in different feature distributions. To prove this observation, we implement a gradient derivation of inter-class separability and intra-class compactness. Concretely, based on (3), the inter-class separability between $\hat{\mathbf{r}}_i$ and the proxies of jth class, along with the intra-class compactness between $\hat{\mathbf{r}}_i$ and its corresponding proxies can be controlled by former and latter equation in (4)

$$s_{\hat{\mathbf{r}}_i,j} \triangleq \sum_{s=1}^{S}\hat{\mathbf{r}}_i\cdot\mathbf{p}_j^s, \qquad s_{\hat{\mathbf{r}}_i,y_i} \triangleq \sum_{s=1}^{S}\hat{\mathbf{r}}_i\cdot\mathbf{p}_{y_i}^s \tag{4}$$

Then, we derive the gradients of L_{mp} with respect to $s_{\hat{\mathbf{r}}_i,j}$ and $s_{\hat{\mathbf{r}}_i,y_i}$. Specifically, the gradient with respect to $s_{\hat{\mathbf{r}}_i,j}$ can be represented as

$$\begin{aligned}\frac{\partial L_{mp}\left(\hat{\mathbf{r}}_i\right)}{\partial s_{\hat{\mathbf{r}}_i,j}} &= \exp\left(s_{\hat{\mathbf{r}}_i,j}\right)\Big/\left[\exp\left(s_{\hat{\mathbf{r}}_i,j}\right)+\sum_{k=1,k\neq j}^{m+n}\exp\left(\sum_{s=1}^{S}\hat{\mathbf{r}}_i\cdot\mathbf{p}_k^s\right)\right]\\ &= 1-\frac{c}{\exp\left(s_{\hat{\mathbf{r}}_i,j}\right)+c}\end{aligned} \tag{5}$$

where $c = \sum_{k=1,k\neq j}^{m+n}\exp\left(\sum_{s=1}^{S}\hat{\mathbf{r}}_\mathbf{i}\cdot\mathbf{p}_\mathbf{k}^\mathbf{s}\right) > 0$ denotes a positive constant irrelevant to the variable $s_{\hat{\mathbf{r}}_i,j}$. From (5) we can conclude that the gradients with respect to the inter-class separability $s_{\hat{\mathbf{r}}_i,j}$ are proportional to the exponential term $\exp(s_{\hat{\mathbf{r}}_i,j})$, and further proportional to $s_{\hat{\mathbf{r}}_i,j}$ due to the monotonicity of the exponential function. In addition, each term in $s_{\hat{\mathbf{r}}_i,j}(\hat{\mathbf{r}}_i\cdot\mathbf{p}_j^s, s = 1,\dots,S)$ owns the common optimal direction (i.e., maximize). Thus, setting larger S brings larger $s_{\hat{\mathbf{r}}_i,j}$ and corresponding larger gradients, which leads to the stronger power of pushing feature $\hat{r}_i$ away from proxies of other classes.

Similarly, based on (5), gradients with respect to $s_{\hat{\mathbf{r}}_i,y_i}$ can be expressed as

$$\frac{\partial L_{mp}\left(\hat{\mathbf{r}}_i\right)}{\partial s_{\hat{\mathbf{r}}_i,y_i}} = -\sum_{k=1,k\neq y_i}^{m+n}\exp\left(\sum_{s=1}^{S}\hat{\mathbf{r}}_i\cdot\mathbf{p}_k^s\right)\Big/\left[\exp\left(s_{\hat{\mathbf{r}}_i,y_i}\right)+\sum_{k=1,k\neq y_i}^{m+n}\exp\left(\sum_{s=1}^{S}\hat{\mathbf{r}}_i\cdot\mathbf{p}_k^s\right)\right] \tag{6}$$

Comparing (5) and (6), we can observe that the magnitude of gradient with respect to the intra-class compactness $s_{\hat{\mathbf{r}}_i,y_i}$ is equal to the sum of gradients with respect to the inter-class separability $s_{\hat{\mathbf{r}}_i,j}$, i.e., $\sum_{j,j\neq y_i}\left|\partial L_{mp}\left(\hat{\mathbf{r}}_i\right)/\partial s_{\hat{\mathbf{r}}_i,j}\right| = \left|\partial L_{mp}\left(\hat{\mathbf{r}}_i\right)/\partial s_{\hat{\mathbf{r}}_i,y_i}\right|$, which urge us to easily conclude that setting larger S also brings stronger power to force feature $\hat{\mathbf{r}}_i$ close to its corresponding proxies.

Moreover, we further observe a key point that S exists an appropriate value. Considering the following extreme case, when $S \to +\infty$, the multi-proxies metric loss L_{mp} can be approximated as (7), which indicates that L_{mp} tends to narrowly focus on the cosine similarity between feature $\hat{\mathbf{r}}_i$ and its nearest proxies of class y_k $(y_k \neq y_i)$. Such tendency traps L_{mp} into the local optimization which absolutely ignores the separability between $\hat{\mathbf{r}}_i$ and other classes except the nearest one. In summary, comparing with Proxy-NCA loss [30], the necessity of setting $S > 1$ in this paper is to enhance the inter-class separability and the intra-class compactness of both old and new classes in feature space, as long as the value of S is appropriate.

$$\begin{aligned}\lim_{S\to+\infty} L_{mp}\left(\hat{\mathbf{r}}_i\right) &= \lim_{S\to+\infty} +\log\left\{1+\sum_{k=1,k\neq y_i}^{m+n}\exp\left[\sum_{s=1}^{S}\left(\hat{\mathbf{r}}_i\cdot\mathbf{p}_k^s-\hat{\mathbf{r}}\cdot\mathbf{p}_{y_i}^s\right)\right]\right\}\\ &= \lim_{S\to+\infty}\max_k\left[\sum_{s=1}^{S}\left(\hat{\mathbf{r}}_i\cdot\mathbf{p}_k^s-\hat{\mathbf{r}}_i\cdot\mathbf{p}_{y_i}^s\right)\right]\end{aligned} \tag{7}$$

Then, the UCL branch locates the upper branch of MPR shown in Fig. 3. The feature $\hat{\mathbf{z}}_{bal}$ of the sample from $\mathbf{D}_{bal}$ is mapped as the final classification logits $\hat{\mathbf{o}}_j \in \mathbb{R}^{m+n}$ for cross entropy loss L_{ce} optimization, which can be formulated as (8). Under the learning of balance sampled data, the weight and bias term of the classification layer tend to be unbiased.

$$L_{ce} = -\sum_{i=1}^{m+n} y_i\log\left(\mathrm{softmax}\left(\hat{\mathbf{o}}_j\right)\right) \tag{8}$$

Overall, the total loss of MPR module is formulated as

$$L_{mpr} = \lambda_1 L_{mp}\left(\mathbf{D}_{rad}\right) + \left(1-\lambda_1\right)L_{ce}\left(\mathbf{D}_{bal}\right) \tag{9}$$

where $\lambda_1 = 1-\left(H/H_{\max}\right)^2$ is a dynamic coefficient related to the current training epoch number H and the total training epoch number $H_{\max}$. Follow the viewpoint that discriminative features are the foundation for learning robust classifiers [31], such coefficient is designed to ensure the model firstly focus on optimizing the backbone CNN based on $\mathbf{D}_{rad}$ and then optimizing the classification layer based on $\mathbf{D}_{bal}$, while avoiding the conflict of different sampled data during training. With the aid of MPR module, the imbalanced learning between old and new classes in both classification space and feature space can be corrected.

3.4. Hybrid knowledge distillation (HKD)

Since only a few old exemplars can attend the incremental update, the discriminative knowledge learned from entire old data will be mostly lost after learning new classes, leading to serious confusion among old classes. To mitigate this issue, we propose the HKD module, which constrains the new model M_t to reproduce same outputs about old classes as the old model M_{t-1}, so that the discriminative knowledge learned by M_{t-1} can be transferred to M_t (as shown in Fig. 4) Specifically, the knowledge in M_{t-1} for discriminating old classes is contained in both the feature space and the classification space. Correspondingly, the proposed HKD contains a feature relation-based distillation (FRD) loss L_{frd} and a response-based distillation (RD) loss L_{rd} to transfer the feature-level old knowledge and the classification-level old knowledge, respectively.

In the feature space of M_{t-1}, old class features are learned to be intra-class compact and inter-class separable. To transfer such knowledge, we first describe it as the pairwise feature topologies matrix. For features $\mathbf{Z}_{rad}$ in the old model, its pairwise topologies matrix $\mathbf{\Gamma}_{t-1} \in \mathbb{R}^{(N/2)\times(N/2)}$ is represented as

$$\mathbf{\Gamma}_{t-1} \triangleq \left[\left\langle\mathbf{z}_{rad}\left(\mathbf{x}_i\right),\mathbf{z}_{rad}\left(\mathbf{x}_1\right)\right\rangle,\dots,\left\langle\mathbf{z}_{rad}\left(\mathbf{x}_i\right),\mathbf{z}_{rad}\left(\mathbf{x}_{N/2}\right)\right\rangle\right], \quad (i = 1,\dots,N/2) \tag{10}$$

where $\langle\cdot\rangle$ represents the cosine similarity. Then, the intra-class compact and inter-class separable knowledge of old class features in M_{t-1} can be described as high similarity among intra-class features and low similarity among inter-class features in $\mathbf{\Gamma}_{t-1}$. For features $\hat{\mathbf{Z}}_{rad}$ in the

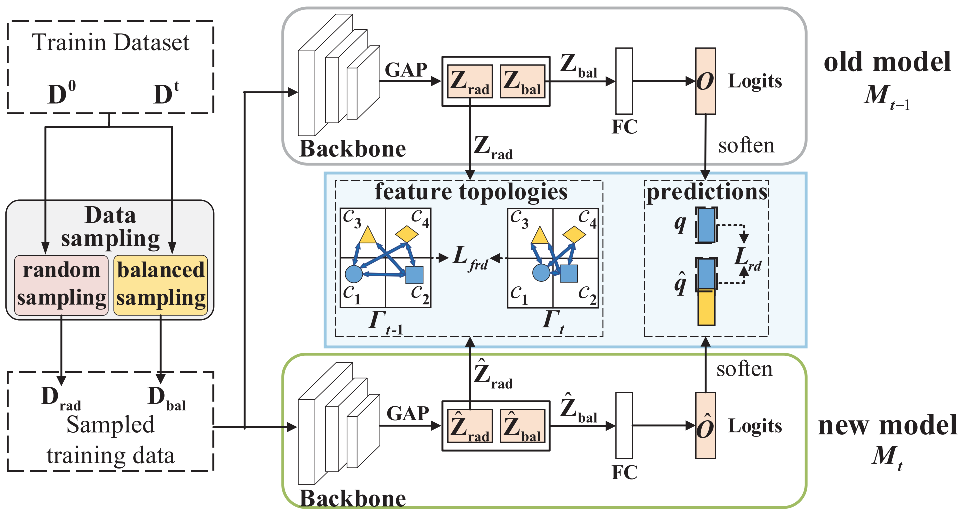

Fig. 4. Hybrid knowledge distillation module schematic diagram.

new model, its pairwise topologies matrix $\mathbf{\Gamma} \in \mathbb{R}^{(N/2)\times(N/2)}$ can be represented as

$$\mathbf{\Gamma}_\mathrm{t} \triangleq \left[\left\langle \hat{\mathbf{z}}_{rad}\left(\mathbf{x}_i\right), \hat{\mathbf{z}}_{rad}\left(\mathbf{x}_1\right)\right\rangle, \ldots, \left\langle \hat{\mathbf{z}}_{rad}\left(\mathbf{x}_i\right), \hat{\mathbf{z}}_{rad}\left(\mathbf{x}_{N/2}\right)\right\rangle\right], \quad (i = 1, \ldots, N/2) \tag{11}$$

Then, the FRD loss can be formulated as follows:

$$L_{frd} = \frac{1}{(N/2)^2} \left\| \mathbf{\Gamma}_{\mathrm{t}\text{ - }1} \odot \mathbf{W}_{mask} - \mathbf{\Gamma}_\mathrm{t} \odot \mathbf{W}_{mask} \right\|_2^2 \tag{12}$$

which achieves the feature-level old knowledge stability by constraining $\mathbf{\Gamma}_t$ in the new model to be consistent with $\mathbf{\Gamma}_{t-1}$ in the old model. In particular, $\mathbf{W}_{mask} \in \mathbb{R}^{(N/2)\times(N/2)}$ is a mask matrix and can be defined as

$$\left(\mathbf{W}_{mask}\right)_{i,j} = \begin{cases} 0, & \text{if } (i,j) \in \mathbf{I}, \\ 1, & \text{else.} \end{cases} \tag{13}$$

Notably, feature topologies $\mathbf{\Gamma}_{t-1}$ computed on $\mathbf{Z}_{rad}$ along with $\mathbf{\Gamma}_t$ computed on $\hat{\mathbf{Z}}_{rad}$ include not only the feature similarity terms among old classes, but also those among new classes. Then, the symbol $\mathbf{I}$ in (13) is a set of 2-dimension indexes, which records the 2-dimension indexes (i,j) of feature similarity terms among new classes in $\mathbf{\Gamma}_{t-1}$ and $\mathbf{\Gamma}_t$. By performing the Hadamard product of $\mathbf{\Gamma}_{\mathrm{t}\text{ - }1} \odot \mathbf{W}_{mask}$ and $\mathbf{\Gamma}_\mathrm{t} \odot \mathbf{W}_{mask}$, the cosine similarity terms among features of new classes in both $\mathbf{\Gamma}_{t-1}$ and $\mathbf{\Gamma}_t$ will be set as 0, and the cosine similarity terms related to old class features will be retained. $\mathbf{W}_{mask}$ is designed on account of the fact that the intra-class high similarity and inter-class low similarity among new classes are not learned in the old model, which helps to extract helpful old knowledge while shielding others.

In the classification space of M_{t-1}, the logits of old data $\mathbf{x}_j$ are learned to possess largest value on true class c_j, denoting the knowledge of correct classification. To transfer such knowledge, we employ the RD loss [16], which restrains the logits on old classes in the old model and new model to be consistent:

$$L_{rd} = \frac{1}{N} \sum_{j=1}^{N} \sum_{i=1}^{m} -\hat{q}_i\left(\mathbf{x}_j\right) \log\left[q_i\left(\mathbf{x}_j\right)\right] \tag{14}$$

where $q_i\left(\mathbf{x}_j\right)$ and $\hat{q}_i\left(\mathbf{x}_j\right)$ denote the ith value of the softened logits $q\left(\mathbf{x}_j\right) \in \mathbb{R}^m$ and $\hat{q}_i\left(\mathbf{x}_j\right) \in \mathbb{R}^{m+n}$, respectively. The softened logit $q_{c_j}\left(\mathbf{x}_j\right) = \exp\left(\mathbf{o}_{c_j}\left(\mathbf{x}_j\right)/\tau\right) / \sum_{l=1}^{m} \exp\left(\mathbf{o}_l\left(\mathbf{x}_j\right)/\tau\right)$ corresponds to the final prediction of the old model for input $\mathbf{x}_j$, while $\hat{q}_i\left(\mathbf{x}_j\right) = \exp\left(\hat{\mathbf{o}}_j\left(\mathbf{x}_j\right)/\tau\right) / \sum_{l=1}^{m} \exp\left(\hat{\mathbf{o}}_l\left(\mathbf{x}_j\right)/\tau\right)$ denotes the final prediction of the new model for $\mathbf{x}_j$. The τ is a temperature scalar.

Overall, our HKD module can retain both the feature and the classification-level old knowledge to fight catastrophic forgetting.

3.5. Integrated objective

Our approach addresses the class imbalance between old and new classes and the catastrophic forgetting towards old classes in CIL from multiple aspects. Combining the losses presented above, the total loss can be defined as

$$L = \lambda_1 \cdot \left[L_{mp}\left(D_{rad}\right) + L_{frd}\left(D_{rad}\right)\right] + \left(1-\lambda_1\right) \cdot \left[\left(1-\lambda_2\right) \cdot L_{ce}\left(D_{bal}\right) + \lambda_2 \cdot L_{rd}\left(D_{bal}\right)\right] \tag{15}$$

where λ_1 and λ_2 are balance coefficients. The coefficient λ_1 has been explained in Section 3.2, while $\lambda_2 = m/m+n$ is designed to force the new model pay more attention to maintaining the classification boundary for old classes via RD loss L_{rd} when the number of old classes m is much larger than that of new classes n. The entire network is jointly optimized in an end-to-end manner.

4. Experiments

4.1. Experimental settings

Datasets. We evaluate our RNKS on two benchmark datasets: (1) CIFAR-100 [32], which contains 100 classes with each class including 600 RGB images of size 32 × 32, 500 images per class for training, and 100 images per class for testing; (2) ILSVRC 2012 [33], which contains 1.28 million training images and 50,000 validation images from 1000 classes.

Benchmark Protocols. As used in benchmark [6], we conduct the experiments according to the following two protocols (1) Datasets settings: in order to simulate CIL processes, each of the above two datasets is randomly divided into batches of data (seen in Eq. (1)) and assigned into multiple learning phases. In detail, for CIFAR-100, we train all 100 classes in batches of 10, 20, or 50 classes (iCIFAR-100) at each learning phase. For ILSVRC 2012, two dataset settings are performed. The first one is using only a random sampled subset of 100 classes, which are trained in batches of 10 (iILSVRC-small). The second one is using all 1000 classes, processed in batches of 100 (iILSVRC-full). (2) Memory size: under the rehearsal framework, the memory size G for old exemplars storation is fixed to 2000 in the experiments on iCIFAR-100 and iILSVRC-small while 20 000 on iILSVRC-full. In each learning phase, G/m exemplars of each old class is stored, where m is the total number of old classes learned before current phase.

Implementation Details. **(1) Implementation details for iCIFAR-100:** For iCIFAR-100, we use 32 layers ResNet [34] as our backbone CNN. The classification layer and the embedding function in MPR are set as a single linear layer. Additionally, different data augmentation strategies are adopted to boost the learning of the two branches in MPR. In detail, 32 × 32 random crop, horizontal flip and random grayscale with probability of 0.2 are used in UCL branch, while the color jitter is further used in SFL branch. The optimizer is momentum SGD with weight decay 1×10^{-4} and momentum 0.9. The mini-batch size N is 256, where half is for random sampled data and remaining is for class-balanced sampled data. The initial learning rate is 0.5 and is divided by 10 after 120 and 160 epochs (200 epochs total). For loss L_{mp}, S is set to 6. For loss L_{rd}, the temperature τ is fixed to 2. The dynamical coefficient $\lambda_1 = 1 - (H/H_{\max})^2$ and the dynamic coefficient $\lambda_2 = m/m+n$. We implement our framework with Pytorch and use an NVIDIA TITAN 2080 Ti GPU to train the network. **(2) Implementation details for ILSVRC:** For ILSVRC, we use 18 layers ResNet [34] as our backbone CNN. The data augmentation is the same as that used in iCIFAR100 except for the random cropping with size 224 × 224. The initial learning rate is also set to 0.5 and divided by 10 every 40 epochs (120 epochs total). The mini-batch size is 128. In addition, for large scale dataset iILSVRC-full, the coefficient λ_1 is set as $\lambda_1 = 1-(H/H_{\max})$ to ensure sufficient learning of the classifier. The other settings are the same as those for the iCIFAR-100.

4.2. Evaluation metrics

To quantitatively evaluate performances of different CIL methods, a widely used average incremental accuracy [6] is employed, which is formulated as:

$$A_{avg} = \frac{1}{T}\sum_{t=2}^{T} a_t \tag{16}$$

where T denotes the total number of learning phases in CIL process, a_t denotes the accuracy computed on the test data of all classes that have been added at tth learning phase (except $t = 1$). In addition, to clearly show the performance after each learning phase, we also provide the accuracy curve, which is organized by $\{a_1, a_2, \dots, a_T\}$. Moreover, following the benchmark [6], the accuracy is computed on top-1 for iCIFAR100 while top-5 for iILSVRC-small and iILSVRC-full.

4.3. Performance comparison

In this section, we compare our RNKS with state-of-the-art methods on iCIFAR100, iILSVRC-small and iILSVRC-full datasets. The state-of-the-art methods include LWF.MC [6], iCaRL-CNN [6], iCaRL-NME [6], EEIL [12], BIC [11], LUCIR [9], WA [1], COIL [18] and OCLvCV [13]. For a fair comparison, all CIL methods are implemented with the same backbone: 32-layers ResNet for iCIFAR100 () and 18-layers ResNet for iILSVRC-small and iILSVRC-full), kept the same memory size (except LWF.MC which only uses distillation strategy without rehearsal), employed the same rehearsal strategy [6] and followed the same assignment of added classes in each learning phase. Moreover, the naive Finetuning (FT) is also implemented, which directly updates the old model with new class data without any incremental strategies.

Table 1 summarizes the average incremental accuracies, the parameter quantity Q and the floating point operations (FLOPs) of all comparison methods and our method on three datasets, where n denotes the numbers of added new classes per learning phase. As shown in Table 1, in the aspect of model complexity, since all listed methods adopt the same backbone network with different training strategies (irrelevant to parameter quantity and forward computation), most of them possess same Q and FLOPs. In particular, BIC adds a bias correction layer and our RNKS adds a feature learning branch, which bring a few augments on Q and FLOPs. In the aspect of model performance, our method achieves the best results on all datasets. In detail, without any incremental strategies, FT constitutes the lower bound of the performance, while LWF.MC improves the performance by employing response-based distillation to prevent catastrophic forgetting. Based on LWF.MC, iCaRL-CNN further leverages rehearsal strategy. The remaining comparison methods integrate various schemes for class imbalance and catastrophic forgetting into iCaRL-CNN. However, most of them narrowly alleviate adverse effects of class imbalance and catastrophic forgetting in classification space, obtaining limited performance improvements. Although LUCIR further impose strong feature-level distillation, its performance is limited by the loss of model flexibility to learn new classes. By reasonably solving the above two issues in feature space further, the average incremental accuracy of our method is 1.23%, 1.93%, 4.05% higher than OCLvCV on iCIFAR-100 (n = 10, 20, 50), slightly higher than WA on iILSVRC-small and iILSVRC-full while outperforms the other methods by a large margin. Considering the complexity and the performance comprehensively, our RNKS achieves dramatic performance improvements with a low cost of adding a single linear layer as the feature learning branch.

In addition, to clearly show the performance after each learning phase, we provide the accuracy curves on the above protocols in Fig. 5. For iCIFAR-100, the accuracy curves of 3 protocols are given in Fig. 5(a)–(c). Fig. 5(a) shows the case of long-term CIL processes that old knowledge tends to be dramatically lost. Our method shows its ability to defy knowledge forgetting by keeping a lower performance degradation throughout such long-term learning. Fig. 5(c) exhibits another extreme case of class imbalance, which preserves 2000 exemplars from 50 old classes to update model with 25 000 new data from 50 new classes. As shown in Fig. 5(c), our method showcases the lowest performance degradation by adequately solving class imbalance. Moreover, the best results are also obtained in the middle case (seen in Fig. 5(b)). For iILSVRC, it can be observed from Fig. 5(d) and (e) that our method obtains the highest accuracy at most learning phases, which demonstrates its validity for various benchmarks.

4.4. Model analysis

4.4.1. Ablation study

In Table 2 we conduct an ablation study by evaluating variations of the proposed method on iCIFAR-100. From Table 2 we can see that the proposed two modules are effective in improving the performance.

Remark 1. For MPR, when only UCL is employed, the performance is worse than the baseline due to the overfitting for balance sampled data.

However, when further introducing SFL learned on random sampled data, the performance obtains 7.43% improvement, validating the remarkable meaning of solving the imbalanced learning in feature space.

Remark 2. For HKD, it should be noticed that when only the distillations are employed, their efforts are not sufficiently apparent since most of the old class data are misclassified into new classes due to class imbalance, while HKD can only retain the discrimination among old classes.

When MPR has been used, our HKD shows significant efforts in boosting performance. In the 6th line, RD improves the performance by 4.66%. In particular, the results in the 5th line is lower than RD, since the discriminative features may still be misclassified by the incorrect classifier. By maintaining correct classifier via RD, the results in the last line prove the importance of further retaining feature-level discriminative knowledge.

The confusion matrices correspond to different variants of our RNKS is further provided in Fig. 6. As seen in Fig. 6(a), the old class data suffers from serious misclassification towards new classes in the baseline. With the help of MPR, such favoritism is effectively alleviated (shown in Fig. 6(b)). Fig. 6(c) exhibits that HKD further decreases the old class confusions caused by catastrophic forgetting.

Table 1
Average incremental accuracy (%), parameter quantity (M) and FLOPs (10^8Mac) of different CIL methods on three datasets.

Method	iCIFAR-100					iILSVRC-small			iILSVRC-full
	Q	FLOPs	A_{avg}			Q	FLOPs	A_{avg}	A_{avg}
	–	–	n = 10	20	50	–	–	10	10
FT	6.67	5.412	19.34	27.94	39.86	11.31	0.745	21.42	18.02
LWF.MC	6.67	5.412	39.71	47.12	52.60	11.31	0.745	60.69	42.50
iCaRL-CNN	6.67	5.412	56.22	58.34	57.45	11.31	0.745	75.45	51.37
iCaRL-NME	6.67	5.412	61.58	63.31	61.97	11.31	0.745	81.80	60.81
EEIL	6.67	5.412	63.56	63.65	60.78	11.31	0.745	88.79	69.36
BIC	6.72	5.413	63.46	65.05	64.89	11.36	0.746	89.50	82.89
LUCIR	6.67	5.412	60.27	62.06	62.73	11.31	0.745	79.70	71.98
WA	6.67	5.412	64.62	66.55	65.07	11.31	0.745	90.15	85.65
COIL	6.67	5.412	64.78	65.08	63.44	11.31	0.745	89.02	83.46
OCLvCV	–	–	67.20	67.64	65.11	–	–	–	–
Ours	6.80	5.414	**68.43**	**69.57**	**69.16**	11.44	0.747	**91.26**	**86.02**

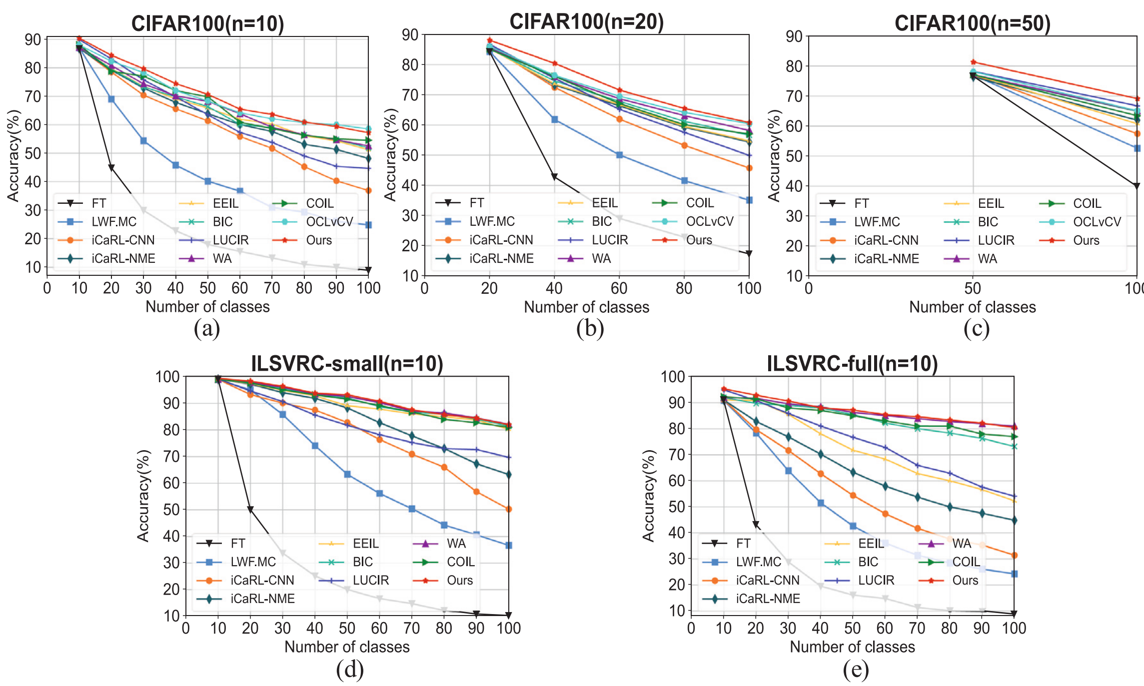

Fig. 5. Accuracy curves of different methods on iCIFAR-100 and iILSVRC.

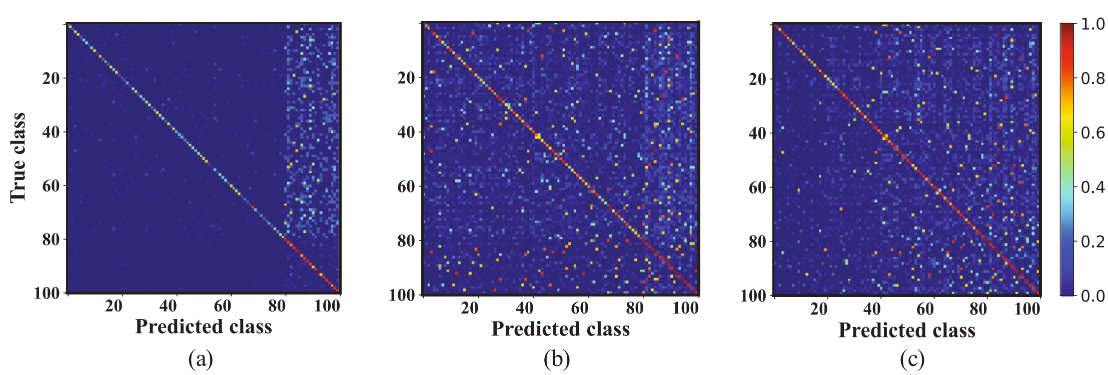

Fig. 6. Confusion matrices of different variants on iCIFAR-100 (with entries being transformed by log(1+x) for better visibility): (a) baseline, (b) baseline + MPR, (c) baseline +MPR+HKD. All of the above confusion matrixes are computed after the last incremental learning phase of iCIFAR-100 (n = 20), where the first 80 classes are old classes and the last 20 classes are newly added classes.

4.4.2. Analysis of MPR

To clearly verify the efforts of MPR in correcting imbalanced learning, the features extracted by the backbone CNN along with weights and biases in the classification layer are visualized in Fig. 7. In the baseline model, most of the new class features are overlapped with old class features (seen in the top of Fig. 7(a)). With the help of SFL in MPR, the above phenomenon can be dramatically alleviated (seen in the bottom of Fig. 7(a)). The visualized analyses of classifier learning are shown in Fig. 7(b) and (c). The top of Fig. 7(b) presents obvious magnitude imbalance between the weights of old and new classes.

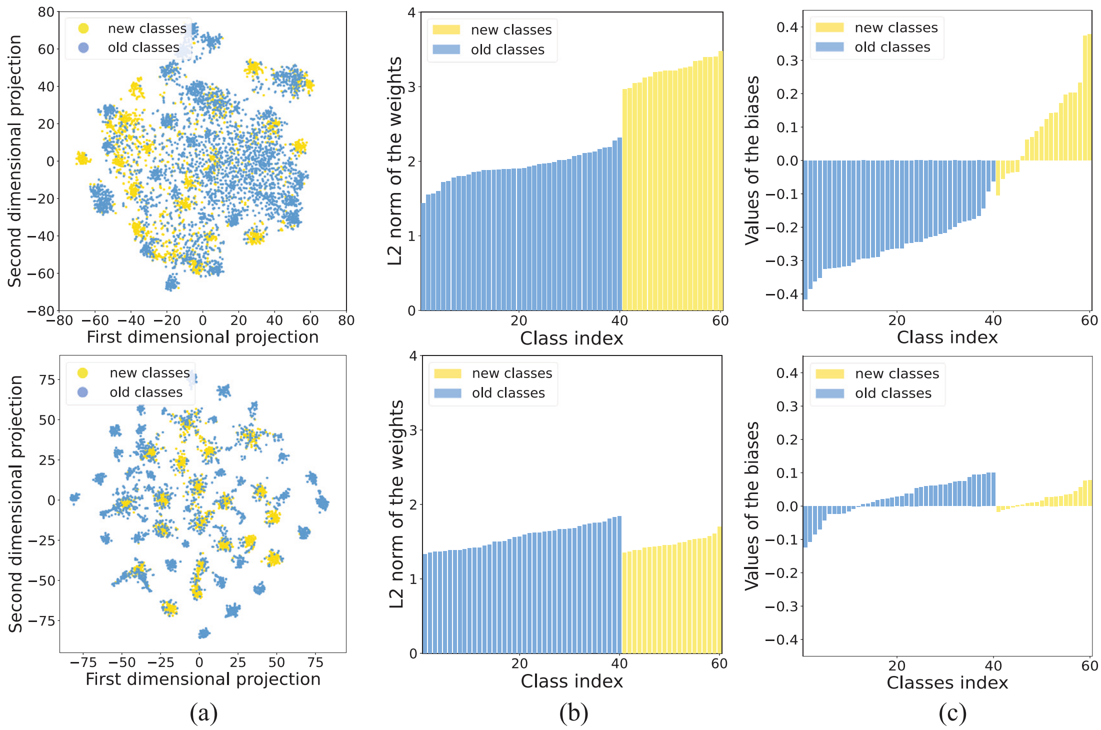

Fig. 7. Visualizations of the features extracted by the backbone CNN along with the weights and biases in the classification layer. In (a), (b), (c), top is the visualizations from the baseline results, while bottom corresponds to the visualizations after using our MPR module. The results come from the 2nd incremental learning phase of iCIFAR-100 (n = 20). (a) 2-D t-SNE visualizations from features of test samples. (b) Visualizations about L2 norms of weights in the classification layer. (c) Visualizations about biases in the classification layer.

Table 2
Ablation study: average incremental accuracy (%) with different modules. Added classes at each learning phase are 20.

Modules		Baseline	Ours					
MPR	UCL	–	✓	✓		✓	✓	✓
	SFL	–		✓		✓	✓	✓
HKD	RD	–			✓		✓	✓
	FRD	–			✓	✓		✓
A_{avg}		56.58	55.26	62.69	59.38	64.13	67.35	**69.57**

Table 3
Comparison of J under different conditions.

Condition	Initial phase	2nd phase without HKD	2nd phase with HKD
J	1.0838	0.6829	1.0188

Similarly, the biases of new classes are mostly positive but those of old classes are negative, as shown in the top of Fig. 7(c). Through the UCL in MPR, the magnitude imbalance can be eliminated (shown in the bottom of Fig. 7(b) and (c)).

4.4.3. Analysis of HKD

To intuitively illustrate the effects of HKD in maintaining feature-level discriminative knowledge, we perform analyses on old class features as below. From the aspect of visualization analyses, the feature 2-d T-SNE along with the pairwise feature similarity matrices are provided in Fig. 8. As seen in Fig. 8(a), the features have been learned to be intra-class compact (high intra-class similarity) and inter-class separable (low inter-class similarity) in the initial phase. However, in the 2nd incremental learning phase, the initial 20 classes become confused due to catastrophic forgetting (shown in Fig. 8(b)). With the help of HKD, the feature discrimination of initial 20 classes can be retained (seen in Fig. 8(c)).

Table 4
Accuracy (%) of initial 20 classes under different conditions.

Condition	Initial phase	2nd phase without HKD	2nd phase with HKD
Accuracy%	88.40	76.35	86.44

From the aspect of numerical analyses, the ratio of inter-class distance to intra-class distance [35] is calculated as $J = trace\left(\mathbf{S}_b\right)/trace\left(\mathbf{S}_w\right)$, where $\mathbf{S}_b$, $\mathbf{S}_w$ represent the inter-class and intra-class scatter matrices and $trace(\cdot)$ is the trace of a matrix. Based on the above definition, the larger the J is, the more discriminative the features are. The results of J under different conditions are listed in Table 3, which shows helpful effects of HKD in retaining feature discrimination.

For analyses of classification-level knowledge maintaining, we perform experiments on predicted normalized logits and the accuracy of old classes as below.

The distribution comparison of predicted logits can be seen in Fig. 9, which shows consistent distribution in Fig. 9(a) and (c) (most of the largest value locates at the true class). However, without HKD, some of the largest value in logits seen in Fig. 9(b) drifts from the true class (marked as white circles).

The comparison of accuracy computed from logits of the initial 20 classes is listed in Table 4. In the initial phase, the accuracy is 88.40%. In the 2nd incremental phase, our HKD significantly improves the performance by 10.09% over the second row and even close to the accuracy in the initial phase.

4.5. Model parameter uncertainty analysis

To evaluate how the hyperparameters in our RNKS affects the learning results of the model parameter, we analysis model parameter uncertainty by performing experiments on variations of the proxies number S in each class, the coefficients λ_2, λ_2 and the memorize size G.

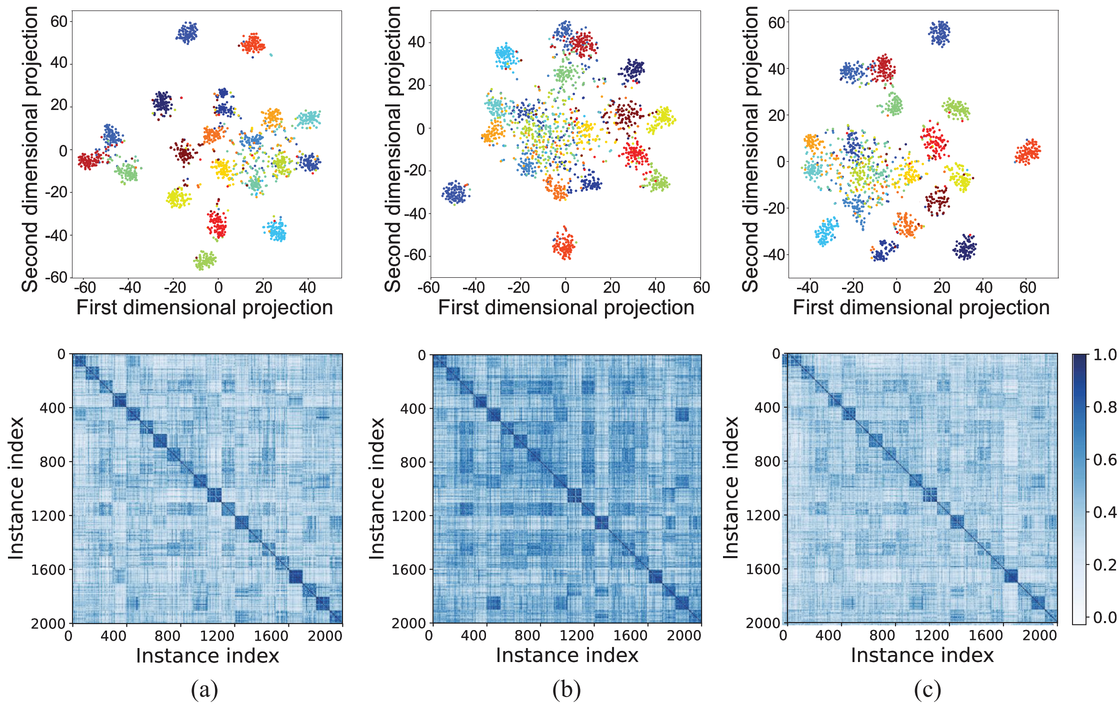

Fig. 8. Visualizations of the test samples belonging to the initial 20 classes in iCIFAR-100 (n = 20). In (a), (b), and (c), the top is the feature 2-D t-SNE and each class is represented by one color, while the bottom is the corresponding pairwise feature similarity matrices sorted by class order and each class contains 100 test samples. (a) Visualizations of the initial phase. (b) Visualizations of the 2nd incremental learning phase without HKD. (c) Visualizations of the 2nd incremental learning phase with HKD. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

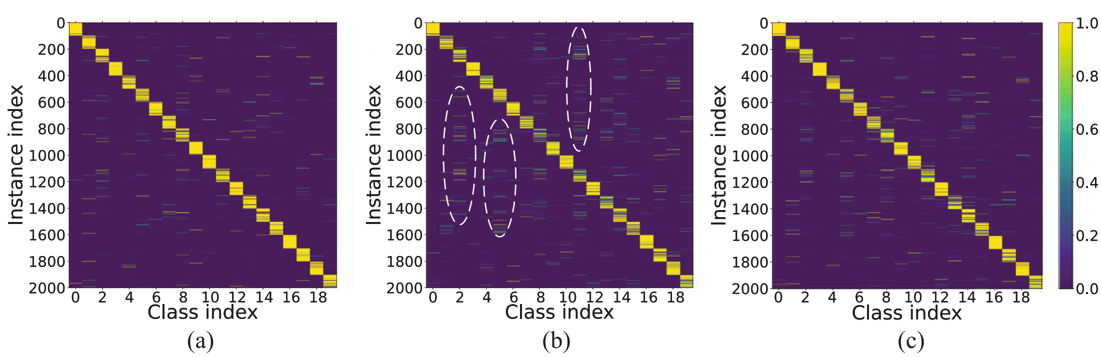

Fig. 9. Visualizations of predicted normalized logits for initial 20 classes in iCIFAR-100 (n = 20). The logits are sorted by class order and each class contains 100 test samples. (a) Visualizations of the initial phase. (b) Visualizations of the 2nd incremental learning phase without HKD. (c) Visualizations of the 2nd incremental learning phase with HKD.

4.5.1. *Effect of proxies number S in multi-proxies metric loss*

First, we visualize the features of training data among old and new classes with different proxy number settings in Fig. 10, and observe that:

(1) The number of proxies in each class are shown to be “1” even under the multiple proxy settings.

Remark 3.1. The optimal directions of multiple proxies in L_{mp} are common, which leads to high similarity among intra-class proxies. However, different settings of proxies have different influences on the optimization.

(2) When employing traditional proxy-NCA (i.e., $S = 1$), there exists some outliers away from their proxies, and the inter-class overlaps also occur. As shown in Fig. 10(b), when S increases, the intra-class distributions of both old and new classes are tighter, and a clearer inter-class separation is also established.

Remark 3.2. Assigning multiple proxies to each class brings stronger constraint to achieve intra-class compactness and inter-class separability.

(3) When S is larger than a logical value, the separation between old and new classes becomes weak, and the proxies drift from its class (seen in Fig. 10(c)).

Remark 3.3. When S is overlarge, the metric loss L_{mp} is limited in the local optimization within a few confused classes while neglects the remaining classes (as derived in (7)).

Fig. 11 intuitively exhibits the effects of different proxy settings on model performance. As shown in Fig. 11, the performance is lowest under $S = 1$ due to its weak constraints. When S increases, the performances are enhanced and are close under $S \in \{5, 6, 8, 10\}$. When $S = 20$, the performance is decreased by the local optimization. Thus, the influences of S on model parameter learning keep stable in an

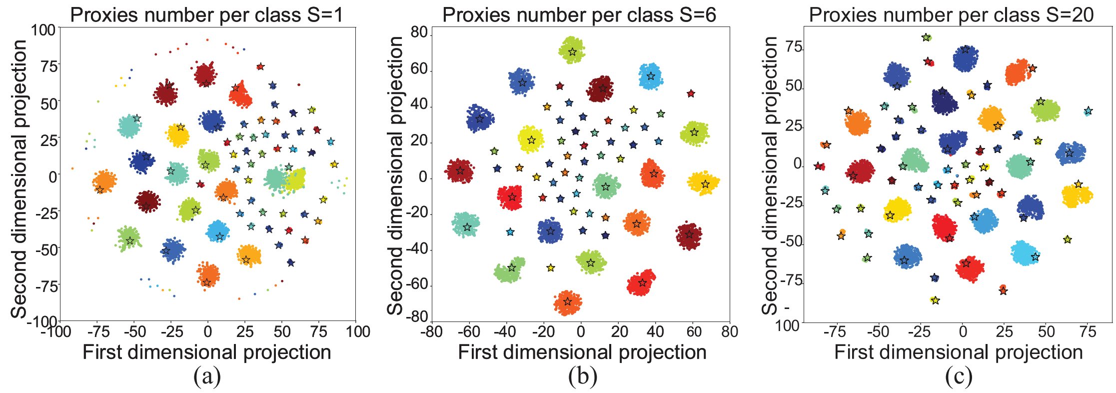

Fig. 10. Visualizations of the features with different proxy settings. The results come from the training data of iCIFAR-100 (n = 20) at 2nd incremental learning phase, where exemplars of each old class are 50 and the data of each new class is 500. The proxies of each class are marked as stars and each class is represented by one color. (a) S = 1. (b) S = 6. (c) S = 20. (d) S = 20. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

appropriate range ($S \in [5, 10]$ for reference), as long as the value is not set to be too small or too large.

4.5.2. Effect of dynamical coefficients

To evaluate influences of coefficient λ_1 in MPR and λ_2 in HKD on model learning, we compare our dynamical coefficient with fixing $\lambda_1, \lambda_2 \in \{1.0, 0.75, 0.5, 0.25, 0\}$. As can be seen in Fig. 12, when fixing $\lambda_1 = 1.0$, the performance is totally destroyed without learning classification layer. As λ_1 gradually decreases to 0, the performance varies due to the different weights on feature and classifier learning. Our dynamical λ_1 achieves the best performance, proving the view point that discriminative features are the foundation for learning robust classifiers. When fixing $\lambda_2 = 1.0$, the performance dramatically decreases without new class learning. As λ_2 gradually decreases to 0, the performance also varies by different weights on old classification boundary retaining and new class learning. Our dynamical λ_2 achieves the best performance, indicating that the above 2 factors should be dynamically adjusted based on the number of old and new classes in different incremental phases. Overall, the model parameter learning is sensitive to the settings of λ_1 and λ_2. Thanks to the automatic determination of them, such sensitiveness can be avoided in applications.

4.5.3. Effect of memorize size G

We conduct the experiments with different memory size. As shown in Fig. 13, all of the CIL methods with rehearsal strategy benefit from a larger memory. Notably, our method shows dominant performance even with the smallest memory size, demonstrating the model stability of our RNKS over other rehearsal CIL methods under storage-limited condition.

5. Summary and discussion

In view of the shortcomings in existing rehearsal-based CIL methods, this paper propose a novel CIL framework RNKS, which mainly contains MPR and HKD to solve class imbalance and catastrophic forgetting, respectively. Differing from previous works in the field which mainly solve the above two issues in classification space, the conspicuous meaning of this work is to reasonably solve them in feature space further, including separability enhancement between old and new classes together with discrimination retaining among old classes. Exhaustive experiments demonstrate that the operations of our work in feature space can dramatically boost CIL performance.

The limitations of this work come down to two points: (1) Data privacy issue: This work assumes a few old exemplars can be saved to attend the CIL process. However, under the data privacy condition, none of the old data can be accessed in CIL, which hinders our MPR from learning the separability between old and new classes. (2) Old

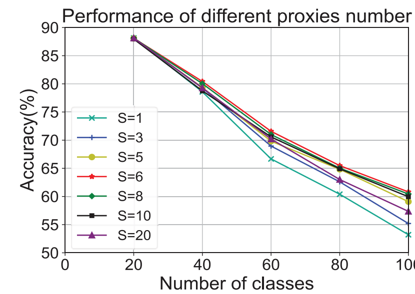

Fig. 11. Performance of different proxies numbers.

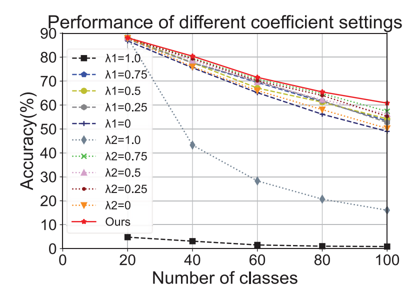

Fig. 12. Performance of different coefficient settings.

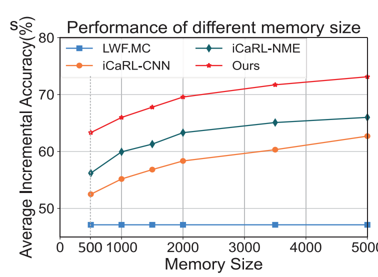

Fig. 13. Performance of different memory size.

class overfitting issue: The model may overfit the distribution of tiny stored old exemplars, limiting the model performance on old classes. In future work, we aim to break the above two limitations. For limitation (1), constructing old class features based on old class proxies to attend model updates may avoid the need for old data storage. For limitation

(2), learning the multimodal distribution of old classes and retaining such distribution during the CIL process can enhance the generalization ability of the model for old classes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgments

This work was partially supported by the National Science Foundation of China (U21B2039, 62201433), the Fundamental Research Funds for the Central Universities (QTZX23067) and the 111 Project.

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